

# Task Manager V5 Full Design Specification

## Purpose

V5 introduces task organization through Priority and Category fields.

The goal is to make tasks more meaningful and prepare the project for future versions such as Search, Filter, and Sort (V6).

V5 is one of the largest versions since V3 because it requires updates to Task, TaskManager, Main, and FileManager.

## Version Goal

Add:

- Priority
- Category

while preserving all V1–V4 functionality including saving/loading, validation, rename, complete, incomplete, and delete.

## Task Class Changes

Current fields:

- name
- completed

New fields:

- priority
- category

Final V5 Task structure:

name  
completed  
priority  
category

## Default Values

New tasks automatically receive:

completed = false  
priority = "Medium"  
category = "Other"

This keeps task creation simple and avoids forcing users to make extra choices.

## Constructors

Keep:

Task(String name)

This constructor sets:

completed = false

priority = Medium  
category = Other

Add/Use:

Task(String name, boolean completed, String priority, String category)

This constructor is used by FileManager when loading tasks from storage.

## Task Methods

Existing:

changeName()  
markComplete()  
markIncomplete()  
getName()  
isCompleted()

New:

setPriority(String priority)  
setCategory(String category)  
getPriority()  
getCategory()

## Priority Design

Priority options:

1. High
2. Medium
3. Low

Users choose numbers from menus.

Main converts:

- 1 -> High
- 2 -> Medium
- 3 -> Low

The Task stores the actual String value.

## Category Design

Predefined categories:

1. School
2. Work
3. Personal
4. Fitness
5. Church
6. Other

Users choose numbers from menus.

Task stores the category as a String.

## TaskManager Changes

Add:

```
boolean setTaskPriority(int taskNumber, String priority)
boolean setTaskCategory(int taskNumber, String category)
```

Return:

```
true = success
false = invalid task number
```

TaskManager remains responsible for finding tasks and delegating changes to Task.

## Architecture

Maintain:

Main

↓

TaskManager

↓

Task

Main handles user input.

TaskManager manages tasks.

Task owns task data.

## Menu Changes

1. Add Task
2. View Tasks
3. Complete Task
4. Delete Task
5. Rename Task
6. Mark Task Incomplete
7. Set Task Priority
8. Set Task Category
9. Exit

## Display Format

Tasks should display as:

Study Java [ ] (High) [School]

Gym Session [X] (Medium) [Fitness]

Format:

Task Name [Status] (Priority) [Category]

## File Format

V3:

Study Java,false

V5:

Study Java,false,High,School

Fields:

name,completed,priority,category

## Backward Compatibility

V5 must still support older V3/V4 save files.

If a task line contains:

Study Java,false

Load it as:

name = Study Java

completed = false

priority = Medium

category = Other

This prevents users from losing existing data.

## Validation Rules

Task names cannot contain commas.

Priority selection:

Invalid choices must re-prompt until valid.

Category selection:

Invalid choices must re-prompt until valid.

If there are no tasks:

Display:

'No tasks available.'

and return to menu.

## User Flow: Set Priority

User selects Set Task Priority

↓

Enter task number

↓

Choose priority

↓

TaskManager.setTaskPriority(...)

↓

Task.setPriority(...)

↓

Success message

## User Flow: Set Category

User selects Set Task Category

↓

Enter task number  
↓  
Choose category  
↓  
TaskManager.setTaskCategory(...)  
↓  
Task.setCategory(...)  
↓  
Success message

## Testing Checklist

Legacy Features:

- ✓ Add Task
- ✓ View Tasks
- ✓ Complete Task
- ✓ Delete Task
- ✓ Rename Task
- ✓ Mark Incomplete
- ✓ Save Tasks
- ✓ Load Tasks

New Features:

- ✓ Set Priority
- ✓ Set Category
- ✓ Priority displays correctly
- ✓ Category displays correctly
- ✓ Priority saves correctly
- ✓ Category saves correctly
- ✓ Priority loads correctly
- ✓ Category loads correctly

Validation:

- ✓ Invalid task numbers
- ✓ Invalid priority choices
- ✓ Invalid category choices
- ✓ Empty task list
- ✓ No commas in task names

## V5 Complete When

- ✓ Priority can be changed
- ✓ Category can be changed
- ✓ Priority displays correctly
- ✓ Category displays correctly
- ✓ Save/load works
- ✓ Backward compatibility works
- ✓ Validation works
- ✓ Old V1–V4 features still work
- ✓ Changes committed to GitHub